Predictive Maintenance using Machine Learning

Machine Failure Prediction using Random Forest.

Presented by — Tanisha Wabale.

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Introduction

Modern industries rely heavily on machines for production. Unexpected machine failures can cause serious problems such as production delays, increased costs, and reduced efficiency.

Traditional maintenance strategies include reactive maintenance and scheduled maintenance. However, these approaches are often inefficient because failures may still occur unexpectedly.

Predictive maintenance uses machine learning techniques to analyze sensor data and predict failures before they occur.

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The Problem Statement

Industries often face **unexpected machine failures**, which lead to:

- Production downtime
- Increased maintenance costs
- Reduced operational efficiency

Traditional maintenance approaches:

- Reactive maintenance (after failure)
- Scheduled maintenance

These approaches are **not always efficient**.

Goal:

Develop a machine learning model to **predict failures in advance**.

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Dataset Overview

The dataset contains machine sensor measurements used to identify failures.

Key Features:

- Torque (Nm)
- Rotational Speed (rpm)
- Tool Wear (min)
- Air Temperature (K)
- Process Temperature (K)

These variables help understand **machine operating conditions**.

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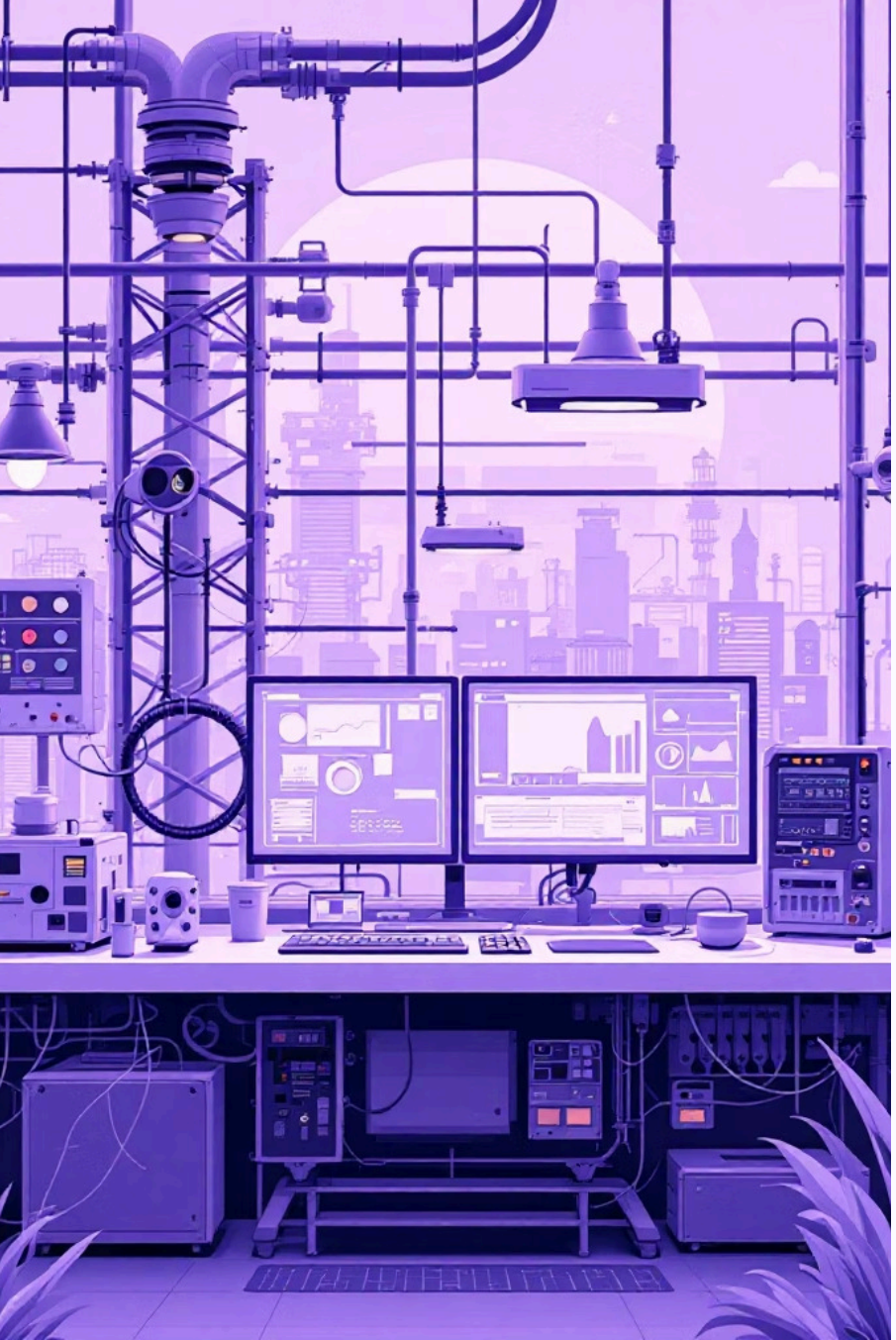

Project Objective

01

Build a machine learning model to predict machine failures.

02

Use sensor data to identify abnormal machine behavior.

03

Support **predictive maintenance strategies.**

04

Reduce operational downtime and maintenance cost.

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Data Preprocessing

Before building the model, several preprocessing steps were performed :

Data Preprocessing

Before building the machine learning model, several preprocessing steps were performed to prepare the dataset and improve model performance.

Feature Preparation

Important features were selected to represent machine operating conditions.

Handling Class Imbalance

SMOTE was applied to balance failure and non-failure data.

Train-Test Split

The dataset was divided into training and testing sets for model development and evaluation.

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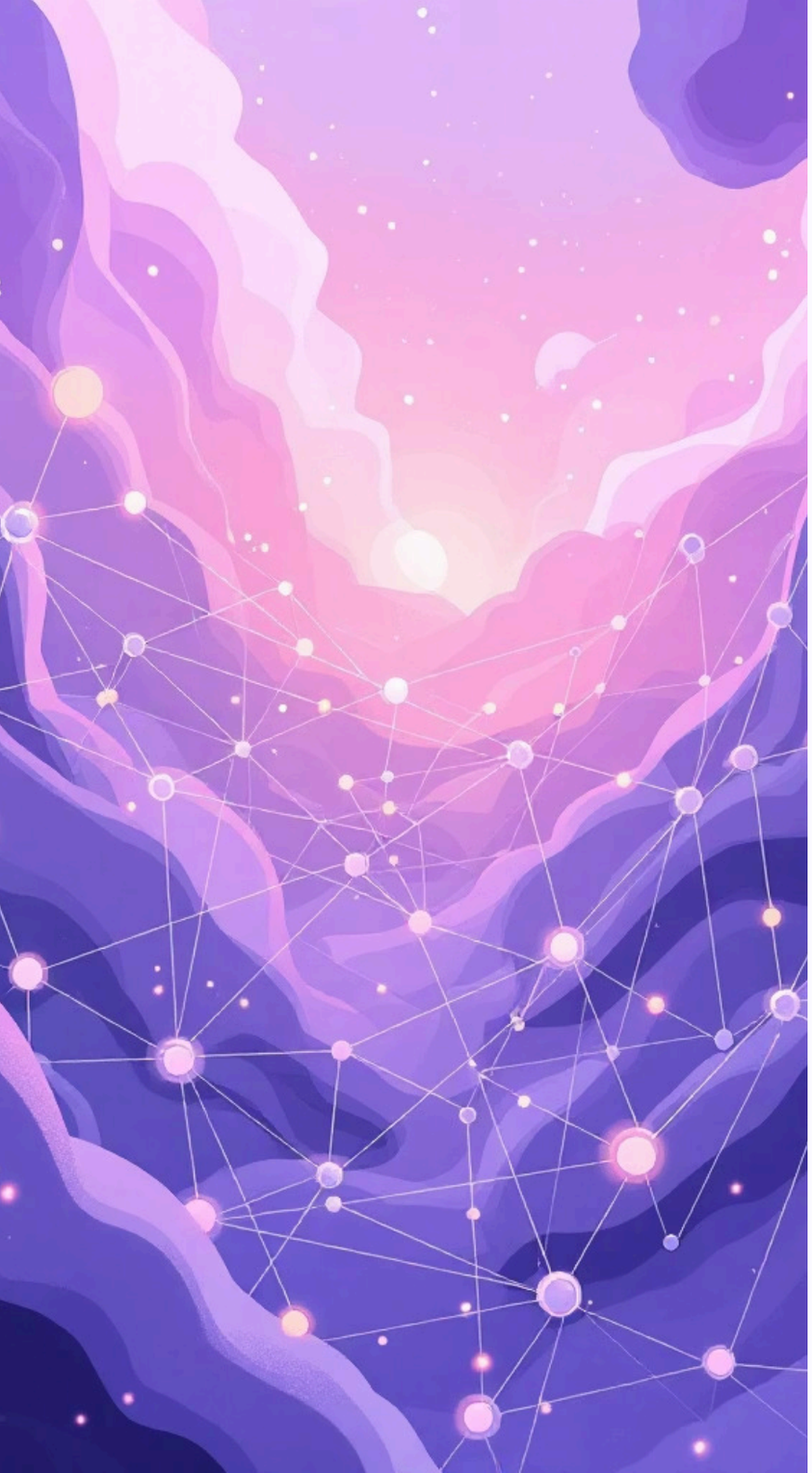

Machine Learning Model & Model Training Process

Machine Learning Model

In this project, the Random Forest classifier was used for machine failure prediction.

Random Forest is an ensemble learning algorithm that combines multiple decision trees to make predictions.

Advantages of Random Forest:

- High accuracy
- Handles structured data effectively
- Reduces overfitting
- Provides feature importance information

Model Training Process

The model training process involved the following steps:

- Splitting the dataset into training and testing sets
- Training the Random Forest model on the training data
- Evaluating the model using testing data
- Measuring model performance using evaluation metrics

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Model Evaluation Metrics

The performance of the model was evaluated using several metrics:

Accuracy

Measures the overall correctness of the model predictions.

Precision

Measures how many predicted failures were actually failures.

Recall

Measures how many actual failures were correctly identified.

F1 Score

Balances precision and recall for better evaluation.

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Model Analysis and Deployment

Feature Importance Visualization

Purpose

- Feature importance analysis was used to identify which variables most influence machine failure predictions.

Key Findings

- **Torque** has the highest importance (~0.33), making it the most influential factor in predicting machine failure.
- **Rotational Speed** is the second most important feature (~0.30).
- **Tool Wear** also plays a significant role (~0.21) in predicting failures.
- **Air Temperature** and **Process Temperature** have relatively lower influence on the model.

Interpretation

- Mechanical stress factors such as **torque and rotational speed** are the strongest indicators of machine failure.
- Monitoring these parameters can help detect potential failures earlier.

Conclusion

- The model successfully identifies the most critical operational parameters that affect machine health.

SHAP Analysis (Model Explainability)

Purpose

- SHAP analysis was used to understand how each feature contributes to the model's predictions.

Key Insights from the SHAP Plot

- The plot shows the **impact of each feature on machine failure prediction**.
- Features such as **torque, rotational speed, and tool wear** have stronger influence on predictions.
- **Higher feature values (red points)** tend to push the prediction toward machine failure.
- **Lower feature values (blue points)** generally push predictions toward normal machine operation.

Interpretation

- Mechanical stress variables play a major role in predicting failures.
- SHAP helps explain **why the model predicts a failure for a specific machine condition**.

Importance

- Improves **model transparency and interpretability**.
- Helps engineers to **trust and understand the machine learning model**.

Stability Analysis

Purpose

- Stability analysis was performed to compare the distribution of features in the **training and testing datasets**.

Key Observations

- The **mean values of all features in train and test datasets are very similar**.
- Features such as **air temperature, process temperature, rotational speed, torque, and tool wear** show almost identical values.
- This indicates **no significant distribution shift between training and testing data**.

Interpretation

- Since both datasets follow similar patterns, the model is **trained on representative data**.
- This improves the model's **generalization ability on new data**.

Conclusion

- The stability analysis confirms that the **model training process is reliable and unbiased**.

Model Deployment Strategy

The trained model can be integrated into a predictive maintenance system.

Workflow

Machine Sensors → Data Collection → ML Model → Failure Prediction → Maintenance Alert

Benefits :

- Early detection of machine issues
- Reduced downtime
- Improved operational efficiency

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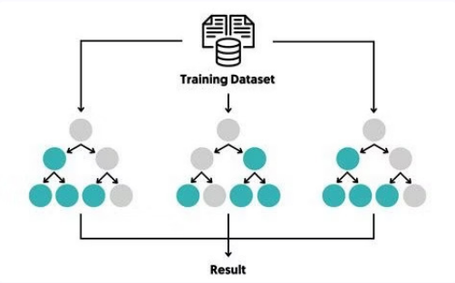
Model comparison table

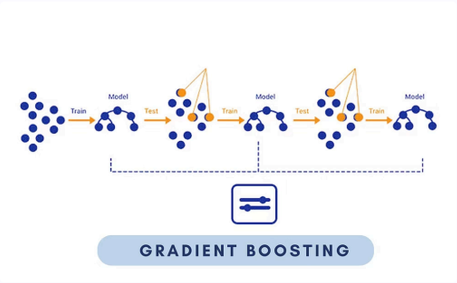


Random Forest

Overall Accuracy: 99.9%

Key Insights

- The **Random Forest model achieved very high accuracy (99.9%)**.
- The model correctly predicted **almost all machine failures**.
- Only **2 failures were misclassified**, showing strong predictive performance.

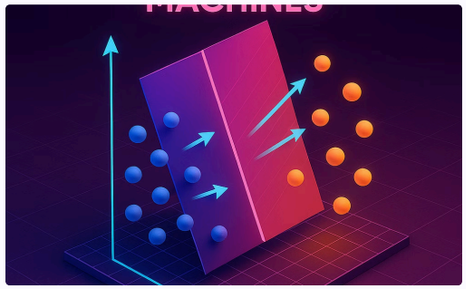


Gradient Boosting

Overall Accuracy: 99.85%

Key Insights

- The **Gradient Boosting model achieved 99.85% accuracy**.
- The model correctly predicted **most machine failures**.
- Only **3 predictions were incorrect**, indicating strong performance.

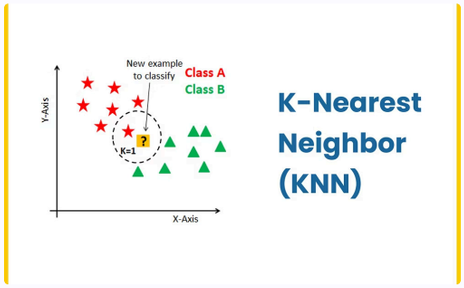


Support Vector Machine (SVM)

Overall Accuracy: 96.95%

Key Insights

- The **SVM model achieved 96.95% accuracy**.
- However, the model **failed to correctly predict machine failure cases**.
- All failure instances were classified as **normal operation**.
- This indicates **poor performance for failure detection**, making SVM unsuitable for this problem.



K-Nearest Neighbors (KNN)

Overall Accuracy: 97.05%

Key Insights

- The **KNN model achieved 97.05% accuracy**.
- It correctly predicted most **normal machine operations**.
- However, **failure detection is weak** with very low recall (0.13).
- Many failure cases were misclassified as normal.

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Case Study Outcome : Predictive Maintenance using Machine Learning

1 The Challenge

Predicting **machine failures using sensor data** to reduce unexpected breakdowns, minimize downtime, and improve maintenance efficiency in industrial systems.

2 The Approach

Applied **machine learning models** such as **SVM, KNN, Gradient Boosting, and Random Forest** on machine sensor data.
Used **feature importance, SHAP analysis, and stability analysis** to interpret model predictions and ensure reliability.

3 The Outcome

- **Random Forest achieved the best performance (99.9% accuracy).**
- Key factors influencing failures were **torque, rotational speed, and tool wear**.
- The model enables **early failure detection and better maintenance planning**, reducing operational costs and downtime.

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Conclusion

- Machine learning models were successfully applied to **predict machine failures using sensor data**.
- Several algorithms were tested, including **SVM, KNN, Gradient Boosting, and Random Forest**.
- Among all models, **Random Forest achieved the highest performance with 99.9% accuracy**, making it the most reliable model for failure prediction.
- **Gradient Boosting** also showed strong performance, while **KNN and SVM performed poorly in detecting failure cases**.
- Feature importance analysis revealed that **torque, rotational speed, and tool wear** are the most influential factors affecting machine failure.
- **SHAP analysis improved model interpretability**, explaining how different features contribute to predictions.
- Stability analysis showed that **training and testing datasets follow similar distributions**, indicating that the model is stable and can generalize well to new data.
- Overall, the developed model demonstrates that **machine learning can effectively support predictive maintenance and reduce unexpected machine downtime**.

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Conclusion & Future Work

Conclusion

- Machine learning can successfully predict machine failures.
- Important operational factors were identified using feature importance and SHAP.
- Predictive maintenance can reduce downtime and maintenance costs.

Future Improvements

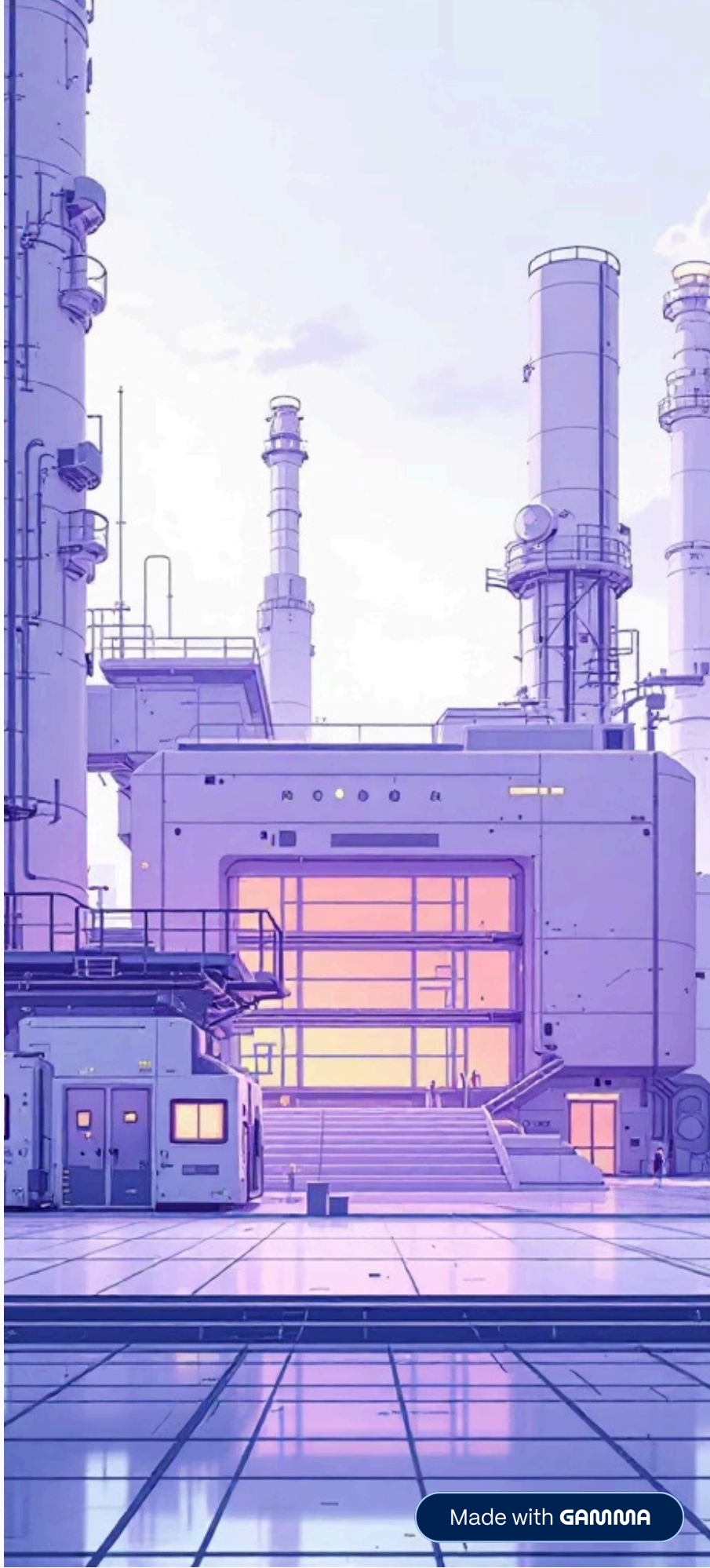
- Integrate real-time IoT (Internet of Things) sensor data
- Use advanced deep learning models
- Develop monitoring dashboards

Predictive maintenance using machine learning enables industries to detect failures early, reduce operational costs, and improve system reliability.

Project Impact

- Reduced downtime
- Lower maintenance costs
- Improved operational efficiency

Thank You.



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